Question 1 (5 marks):

This question is a simplified version of assignment #1.

A file named *density1.data* (you may assume redirection is being used if you wish) contains a survey of soil density values for a terrain destined to hold a high-rise condominium building. You need to use that file to calculate the average density for that site and make a decision before building. If the average is below 1.5, the site is unusable (building there will not be safe), otherwise it is safe to build.

Your program must display the average density with three decimal digits precision and the building recommendation (safe or unsafe).

A soil density file contains an unknown quantity of double values. Assume that a file contains at least one value in it. Assume all values are valid doubles.

```
#include <stdio.h>
int main
{
  ~~double value, ave, sum, density;~~
  double value, ave=0, sum=0, density;
  int count;
  FILE *density;
  density = fopen ("density1.data", "r");
  while (fscanf ("density", "%lf", &value) != EOF).
  {
      sum = sum + value;
      count = count + 1;
  }
  ave = sum/count;
  if (average <1.5)
      printf ("The average density is %.3lf, site is unsafe", ave);
  else
      printf ("The average density is %.3 lf, site is safe", ave);
  fclose (density);
  return 0;
}
```

-0.5

```
#include <stdio.h>
#include <stdio.h>
int main()
{
    int num1, num2, num3, num4, sum1, sum2, sum11
    double ~~sum2~~ sum3
    Printf("Enter 4 integer values");
    scanf("%d,%d,%d,%d",&num1, &num2, &num3, &num4);
    if


    sum2 = pow(num1,2) + pow(num2,2) + pow(num3,2) + pow(num4,2)
    sum1 = num1*num1 + num2*num2 + num3*num3 + num4*num4
    sum3 = (sqrt(sum2))/sum11
    sum11 = n1 + n2 + n3 + n4

    ~~while~~
    if num1 <27 && num1 >7
        sum1 = sum1 + num1;


for (i=0; i<4; i++)
{   printf("enter num");
    scanf("%d", num);
  if (num >7 && num <27)
  {  sum = sum+num;
  else sum = sum  }
  sum1 = sum1 + num;
  sum2 = sum2 + pow(num,2);
  sum3 = (sqrt(sum2)) / sum1;
}
```

```
#include <stdio.h>
#include <math.h>
int main
{
    int num,i, sumtotal, sum a, sum b;
    double quotient;
    for (i=0; i<8; i++)
    { Printf("enter a number");
      Scanf("%d",&num);
      if (num>7 && num<27)
          { sum= sum+ num;
          }
          else
          sum=sum; }
          sum1 = sum1+n;
          sum b = sum2 + pow(num,2);
quotient ~~sum~~ c = (sqrt(sum b))/sum1;
}
Printf("sum of all the numbers between 7
```

Question 2 (5 marks):

This program you may recall from lab #3 with small changes.
Write a C program that will ask the user for 4 integer values and then produce the following output:
a. The sum of all the numbers (but only the ones between 7 and 27 inclusive).
b. The sum of the squares of the four numbers.
c. The quotient (with 2 digits decimal precision) of the square root of the sum of the squares divided by the sum of all the numbers.

Prompt for all values. Use meaningful variable names. Clearly label all output values.

```
#include <stdio.h>
#include <math.h>
int main ()
{
    int num, sumtotal, suma, sumb;
    double quotient;
    for (i=0; i<3; i++)
      { printf("enter an integer");
        scanf("%d", &num);
        sumtotal= sumtotal + num;
        sumb = sumb + pow(num,2);
        quotient = (sqrt(sumb)) / sumtotal;
        if (num>7 && num<27)
          { suma = suma + num;
          }
          suma= suma;
      }
    printf("The sum of all numbers between 7 and 27 is %d", suma);
    printf("The sum of all the squares is %d", sumb);
    print("The quotient of the sum of the squares devided by total sum is %.2 lf", quotient);
    return 0;
}
```

```
double
 formula ( double n)
 {
    y = (n*(n+1)*(2*n+1))/6;
    return (y);
 }

int main
{      double y, n;
printf("please enter n");
scanf("%lf", &n);
 formula(n);
 printf("%.2lf", y);
 return(0);
}
```

Question 3 (5 marks):

The answer to this question should be a C program that must be structured as a main function and a helper (user-defined) function named *formula*.

Complete the helper (user-defined) function named ***formula*** that implements the following formula:

$$y = \frac{n\ (n+1)\ (2n+1)}{6}$$

Next, write the main function. It should prompt the user for a double number *n*. The main function then calls the *formula* function and displays the value of y (with two decimal digits precision). It is forbidden to use pointers in this question.

```
#include <stdio.h>
#include <math.h>

double
formula ( double n )                    )
{
  double y;
  y = (n*(n+1)*(2*n+1))/6
  return (y);


}
/* write the main program here */
int main
{
   double y, n;
   printf("please enter value for n");
   scanf ("%.1f", &n);
   formula (n);
   printf ("%.2lf", y);
   return (0);
}

/* end of main program */
```

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Question #4 (5 marks):

Complete the following C program that changes the value of an integer variable from 24 to 25.
Read the comments for directions on what to write.

Write only ONE statement per blank line.
The numbers are printed alone (no text) and on two separate lines.
The parameter in the *change* function **must be a pointer**.
There is NO RETURN STATEMENT in the *change* function.
It is forbidden to do computations or calls in the printf statements.

```
#include <stdio.h>

void
change (____int *a____) /* write the parameter */
{
    *a= *a+1                    -1          /* change the value to 25 */
}

int
main (void)
{

   int  a = 24;

   printf ("%d", a                    ); /* print original a */

   /* call the change function to change a from 24 to 25 */
   change (&a); (&a)   -2½

   printf ("%d", a                    ); /* print new a */
   return (0);
}
```

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